Intelligent Reflecting Surface Assisted Anti-Jamming Communications in THz Wideband Systems: A Fuzzy WoLF-PHC Learning Approach with Hybrid Beamforming

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***Abstract*—Intelligent reflecting surface (IRS) assisted wireless communication is a promising paradigm for enhancing spectral efficiency and security against malicious jamming. This paper investigates the joint optimization of transmit power allocation and IRS reflecting beamforming in a wideband terahertz (THz) system operating at 100 GHz with 10 GHz bandwidth, where a smart multi-antenna jammer attacks legitimate user transmissions. We propose a hybrid reinforcement learning (RL) and alternating optimization (AO) framework: RL handles discrete power allocation decisions while AO optimizes continuous IRS phase shifts via closed-form updates. To cope with the dynamic and unknown jamming environment, we employ a fuzzy win-or-learn-fast policy hill-climbing (WoLF-PHC) algorithm that combines fuzzy state aggregation with adaptive mixed-strategy learning. The THz extension incorporates sub-connected hybrid beamforming at the base station, sub-array phase-dependent partial-connection (SPDP) at the IRS for wideband beam squint compensation, and OFDM signaling across 64 subcarriers. Simulation results demonstrate that the proposed fuzzy WoLF-PHC approach achieves a system rate of 11.46 bits/s/Hz and 78.6% SINR protection level, outperforming classical Q-learning (7.13 bits/s/Hz, 38.8%), deep Q-network (7.68 bits/s/Hz, 41.9%), and the AO baseline (8.01 bits/s/Hz, 44.8%). Furthermore, sweep analyses over transmit power, IRS size, SINR threshold, and jammer power confirm that WoLF-PHC consistently dominates all baselines, with particularly strong robustness under increasing jammer power.**

***Index Terms*—Anti-jamming, intelligent reflecting surface, terahertz communications, reinforcement learning, beamforming, beam squint, WoLF-PHC.**

I. Introduction

DUE to the inherent broadcast nature of wireless channels, wireless transmissions are vulnerable to malicious jamming attacks where jammers intentionally emit interfering signals to degrade communication performance [1], [2]. Defending against such attacks is a critical security challenge in both current fifth-generation (5G) and future sixth-generation (6G) wireless networks.

Traditional anti-jamming techniques include frequency hopping [3], power control [4], cooperative relaying [5], and beamforming-based approaches. However, these methods face limitations: frequency hopping requires extra spectrum, power control is energy-consuming, and active relays incur high hardware cost and complexity.

Intelligent reflecting surfaces (IRSs) have emerged as a cost-effective paradigm for enhancing both spectral efficiency and physical-layer security [6], [7]. An IRS comprises a large number of passive reflecting elements, each of which adjusts its reflection phase to constructively combine desired signals while mitigating interference [8]. The passive nature of IRS eliminates the need for active radio frequency (RF) chains, making it ideal for deployment in jamming scenarios where power resources are contested.

Yang et al. [1] pioneered the application of IRS to anti-jamming communications by proposing a fuzzy WoLF-PHC learning approach that jointly optimizes power allocation and IRS phase shifts against a smart jammer. While groundbreaking, their work considered narrowband sub-6 GHz operation with a uniform linear array (ULA) at the base station (BS).

A. Motivation and Contributions

The migration to terahertz (THz) bands (0.1–10 THz) promises massive bandwidths but introduces unique challenges: severe path loss, molecular absorption, beam squint across wideband OFDM subcarriers, and the need for large antenna arrays with hybrid analog-digital beamforming [9], [10]. These challenges are further exacerbated in adversarial environments where a smart jammer exploits the directional nature of THz links.

In this paper, we extend the IRS-assisted anti-jamming framework to wideband THz systems and make the following contributions:

- We design a wideband THz IRS-assisted anti-jamming system operating at $f_c = 100\,\text{GHz}$ with $B = 10\,\text{GHz}$ bandwidth, featuring a 256-element IRS with sub-array phase-dependent partial-connection (SPDP) architecture for beam squint compensation, a 64-antenna BS with sub-connected hybrid beamforming, and OFDM signaling with 64 subcarriers.
- We propose a hybrid RL + AO framework where reinforcement learning handles discrete power allocation (42 actions) and alternating optimization solves continuous IRS phase shifts via closed-form updates, achieving tractable optimization in the high-dimensional action space.

- We employ a fuzzy WoLF-PHC algorithm with 3-feature state compression and 27-component fuzzy membership that provides smooth generalization across states and implements an adaptive mixed-strategy policy that is inherently unpredictable to the smart jammer.
- We provide comprehensive simulation results including parameter sweeps over transmit power ($P_{\max}$), IRS size (N_{RIS}), SINR target ($\gamma_{\min}$), and jammer power (P_J), along with SINR cumulative distribution function (CDF) analysis and runtime comparisons, demonstrating consistent superiority of the proposed approach.

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III details the THz-specific channel model and wideband beamforming design. Section IV describes the RL framework and the proposed fuzzy WoLF-PHC algorithm. Section V presents simulation results, and Section VI concludes the paper.

Notation: Boldface lowercase and uppercase letters denote vectors and matrices, respectively. $(\cdot)^H$, $(\cdot)^*$, and $(\cdot)^T$ denote conjugate transpose, conjugate, and transpose. $|\cdot|$ and $\|\cdot\|$ denote absolute value and Euclidean norm. $\mathbb{C}^{M\times N}$ is the space of $M \times N$ complex matrices. $\mathbb{E}[\cdot]$ is statistical expectation. $\mathrm{diag}(\cdot)$ forms a diagonal matrix. $\mathbf{1}\{\cdot\}$ is the indicator function.

II. System Model and Problem Formulation

A. System Description

We consider a downlink IRS-assisted communication system (Fig. 1) consisting of a BS equipped with N antennas and N_{RF} RF chains serving K single-antenna legitimate users. An IRS with $M = N_1 \times N_2$ reflecting elements is deployed to enhance communication links. A smart multi-antenna jammer with N_J antennas attempts to disrupt the legitimate transmissions.

The system parameters for the THz wideband configuration are:

- BS: $N = 64$ ULA antennas, $N_{\text{RF}} = 8$ RF chains
- IRS: $M = 256$ elements (16×16 UPA), $Q = 8 \times 8$ sub-arrays
- Users: $K = 4$ single-antenna UEs
- Jammer: $N_J = 2$ antennas (portable THz jammer)
- OFDM: $M_{\text{sc}} = 64$ subcarriers, $B = 10\,\text{GHz}$, $f_c = 100\,\text{GHz}$

B. Signal Model

Let $\mathbf{G}[m] \in \mathbb{C}^{M\times N}$ denote the BS-IRS channel at subcarrier m, $\mathbf{g}_{ru,k}[m] \in \mathbb{C}^M$ the IRS-user-k channel, $\mathbf{g}_{bu,k}[m] \in \mathbb{C}^N$ the direct BS-user-k channel, and $\mathbf{h}_{J,k} \in \mathbb{C}^{N_J}$ the jammer-user-k channel. The IRS reflection matrix at subcarrier m is $\mathbf{\Phi}[m] = \mathrm{diag}(e^{j\theta_1[m]}, \ldots, e^{j\theta_M[m]})$.

The received signal at user k on subcarrier m is

$$\begin{aligned} y_k[m] = &\underbrace{\left(\mathbf{g}_{ru,k}^H[m]\mathbf{\Phi}[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m]\right)\mathbf{w}_k[m]\sqrt{P_k}\,s_k}_{\text{desired signal}} \\ &+ \underbrace{\sum_{i\neq k}\left(\mathbf{g}_{ru,k}^H[m]\mathbf{\Phi}[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m]\right)\mathbf{w}_i[m]\sqrt{P_i}\,s_i}_{\text{inter-user interference}} \\ &+ \underbrace{\sqrt{P_{J,k}}\,\mathbf{h}_{J,k}^H\mathbf{z}_k\,j_k}_{\text{jamming}} + n_k \end{aligned} \tag{1}$$

where $\mathbf{w}_k[m] \in \mathbb{C}^N$ is the beamforming vector, P_k is the allocated power, s_k is the unit-power data symbol, $P_{J,k}$ is the jamming power, $\mathbf{z}_k$ is the jammer precoder, j_k is the jamming symbol, and $n_k \sim \mathcal{CN}(0,\sigma^2)$ is additive white Gaussian noise with $\sigma^2 = k_B T B/M_{\text{sc}} + F_{\text{NF}}$ where k_B is Boltzmann's constant, $T = 290$ K, and $F_{\text{NF}} = 10\,\text{dB}$ is the noise figure.

C. SINR and System Rate

The effective channel for user k at subcarrier m is

$$\mathbf{h}_{\text{eff},k}[m] = \mathbf{G}^H[m]\mathbf{\Phi}^H[m]\mathbf{g}_{ru,k}[m] + \mathbf{g}_{bu,k}[m]. \tag{2}$$

The received SINR at user k on subcarrier m is

$$\text{SINR}_k[m] = \frac{P_k|\mathbf{h}_{\text{eff},k}^H[m]\mathbf{w}_k[m]|^2}{\sum_{i\neq k} P_i|\mathbf{h}_{\text{eff},k}^H[m]\mathbf{w}_i[m]|^2 + P_{J,k}|\mathbf{h}_{J,k}^H\mathbf{z}_k|^2 + \sigma^2}. \tag{3}$$

The per-user rate averaged over subcarriers is

$$R_k = \frac{1}{M_{\text{sc}}}\sum_{m=1}^{M_{\text{sc}}}\log_2(1 + \text{SINR}_k[m]) \tag{4}$$

and the system sum-rate is $R_{\text{sum}} = \sum_{k=1}^K R_k$.

The SINR protection level measures quality-of-service (QoS) satisfaction:

$$\eta_{\text{prot}} = \frac{1}{K\cdot M_{\text{sc}}}\sum_{k=1}^{K}\sum_{m=1}^{M_{\text{sc}}}\mathbf{1}\{\text{SINR}_k[m] \geq \gamma_{\min}\} \tag{5}$$

where $\gamma_{\min}$ is the minimum SINR threshold.

D. Problem Formulation

The joint optimization of power allocation and IRS beamforming is formulated as

$$\max_{\{P_k\},\{\mathbf{\Phi}[m]\}} \quad R_{\text{sum}} \tag{6a}$$

$$\text{s.t.} \quad \sum_{k=1}^K P_k \leq P_{\max} \tag{6b}$$

$$\text{SINR}_k[m] \geq \gamma_{\min},\ \forall k, m \tag{6c}$$

$$|\Phi_{m,m}[n]| = 1,\ \forall m, n \tag{6d}$$

where (6b) is the total power constraint, (6c) is the per-user QoS constraint, and (6d) is the unit-modulus constraint on IRS elements.

Problem (6a)–(6d) is non-convex and NP-hard due to the coupled optimization variables and the unit-modulus constraint. Furthermore, the jammer's behavior is dynamic and

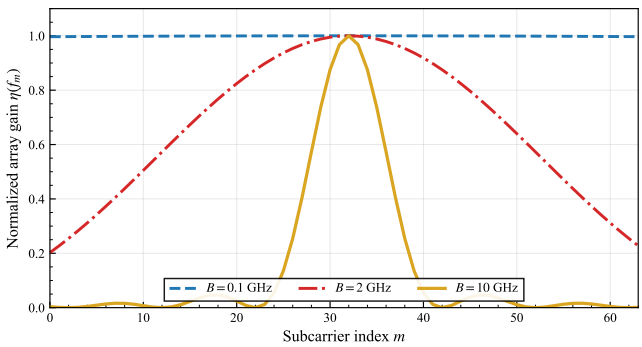

Fig. 1. Beam squint effect: normalized array gain $\eta(f_m)$ across 64 OFDM subcarriers for bandwidths $B = 0.1$, 2, and 10 GHz at $f_c = 100$ GHz. Wider bandwidth causes severe gain degradation at edge subcarriers, motivating the SPDP-RIS architecture.

unknown, making model-based optimization impractical. We therefore model this as a reinforcement learning problem with AO-assisted continuous optimization.

III. THz Channel Model and Wideband Beamforming

A. THz Channel Model

At THz frequencies, the channel exhibits frequency-selective behavior due to molecular absorption and beam squint. The BS-IRS channel at subcarrier m follows a Saleh-Valenzuela model [9]:

$$\mathbf{G}[m] = \sqrt{\frac{NM}{L_{\text{path}}}} \sum_{\ell=1}^{L_{\text{path}}} \alpha_\ell(f_m)\, \mathbf{a}_{\text{RIS}}(\phi_\ell, f_m) \mathbf{a}_{\text{BS}}^H(\psi_\ell, f_m) \quad (7)$$

where L_{path} is the number of paths, $\alpha_\ell(f_m)$ is the complex path gain at subcarrier frequency $f_m = f_c + (m - M_{\text{sc}}/2)\Delta f$ with $\Delta f = B/M_{\text{sc}}$, and $\mathbf{a}_{\text{RIS}}$, $\mathbf{a}_{\text{BS}}$ are frequency-dependent array response vectors.

B. Path Loss and Molecular Absorption

The THz path loss at frequency f and distance d is

$$\text{PL}(f, d) = \left(\frac{4\pi f d}{c}\right)^2 e^{\kappa_{\text{abs}}(f) d} \quad (8)$$

where c is the speed of light and $\kappa_{\text{abs}}(f)$ is the frequency-dependent molecular absorption coefficient, which creates spectral windows and opacity regions in the THz band.

C. Beam Squint in Wideband THz Systems

A critical challenge in wideband THz systems is *beam squint*: the spatial direction of a phased-array beam shifts with frequency. For a ULA with inter-element spacing $d_s = \lambda_c/2$ steered to angle θ_0 at the center frequency, the normalized array gain at subcarrier m is

$$\eta(f_m) = \frac{1}{N} \left| \sum_{n=0}^{N-1} e^{j2\pi n d_s \sin\theta_0 \left(\frac{f_m}{f_c} - 1\right)/\lambda_c} \right|^2. \quad (9)$$

As shown in Fig. 1, for $B = 10$ GHz at $f_c = 100$ GHz (fractional bandwidth $\beta = B/f_c = 10\%$), edge subcarriers suffer severe gain loss. This motivates the SPDP-RIS architecture.

D. SPDP-RIS Architecture

To combat beam squint at the IRS, we adopt the sub-array phase-dependent partial-connection (SPDP) architecture [9]. The M RIS elements are partitioned into $Q = Q_1 \times Q_2$ sub-arrays, each containing $M_s = M/Q$ elements. Each sub-array applies a common true-time-delay (TTD) and per-element phase shifter, enabling frequency-dependent compensation:

$$\theta_{q,i}[m] = \theta_{q,i}^{\text{base}} + 2\pi f_m \tau_q \quad (10)$$

where $\theta_{q,i}^{\text{base}}$ is the base phase of element i in sub-array q, and τ_q is the TTD value for sub-array q.

E. Hybrid Beamforming at the BS

The BS employs a sub-connected hybrid architecture with $N_{\text{RF}} = 8$ RF chains, each driving $N/N_{\text{RF}} = 8$ antennas. The beamforming vector decomposes as

$$\mathbf{w}_k[m] = \mathbf{F}_{\text{RF}} \mathbf{f}_{BB,k}[m] \quad (11)$$

where $\mathbf{F}_{\text{RF}} \in \mathbb{C}^{N \times N_{\text{RF}}}$ is the analog precoder (block-diagonal, constant across subcarriers) and $\mathbf{f}_{BB,k}[m] \in \mathbb{C}^{N_{\text{RF}}}$ is the digital baseband precoder per subcarrier.

The digital precoder uses a max-SINR (MVDR) design. For each user k at subcarrier m, the interference-plus-noise covariance matrix is

$$\mathbf{R}_k[m] = \sigma^2 \mathbf{I} + \sum_{i \neq k} P_i\, \tilde{\mathbf{h}}_i[m] \tilde{\mathbf{h}}_i^H[m] \quad (12)$$

where $\tilde{\mathbf{h}}_k[m] = \mathbf{F}_{\text{RF}}^H \mathbf{h}_{\text{eff},k}[m]$ is the effective channel in the reduced-dimension baseband. The MVDR baseband precoder is

$$\mathbf{f}_{BB,k}[m] = \frac{\mathbf{R}_k^{-1}[m] \tilde{\mathbf{h}}_k[m]}{\|\mathbf{R}_k^{-1}[m] \tilde{\mathbf{h}}_k[m]\|}. \quad (13)$$

IV. Reinforcement Learning Framework

A. Hybrid RL + AO Decomposition

The optimization problem is decomposed into:

1) **RL decision (discrete):** The RL agent selects a power allocation action from a finite set $\mathcal{A}$ of $|\mathcal{A}| = 42$ actions, comprising power fraction indices and distribution modes.
2) **AO optimization (continuous):** Given the power allocation, IRS phase shifts are optimized via closed-form alternating updates.

1) Power Allocation Actions: Each action is a pair (fraction, mode):

- 7 total power fractions of $P_{\max}$: $\{0.2, 0.35, 0.5, 0.65, 0.8, 0.9, 1.0\}$
- 6 distribution modes: equal, channel-proportional, inverse-channel, SINR-deficit, waterfilling, max-min fairness

yielding $7 \times 6 = 42$ discrete actions.

2) IRS Phase Optimization via AO: Given power $\{P_k\}$ from the RL agent, IRS phases are updated in closed form. For each element i:

$$c_i = \sum_{k=1}^{K} \sqrt{P_k}\, g_{ru,k}^*[i] \cdot (\mathbf{G}\mathbf{w}_k)[i], \quad \theta_i = -\angle(c_i) \tag{14}$$

which maximizes the sum of desired signal magnitudes. A secondary SINR-deficit-weighted AO iteration is also computed, and the strategy yielding higher sum-rate is selected (6 AO iterations).

B. State Representation

The high-dimensional observation is compressed into a 3-feature state vector $\mathbf{f} = [f_{\text{pj}}, f_{\text{ch}}, f_{\text{sinr}}] \in [0,1]^3$:

Jammer pressure:

$$f_{\text{pj}} = \text{norm}_{[15,40]}\left(0.6\overline{P_J}\,[\,\text{dBm}] + 0.4 \max_k P_{J,k}\,[\,\text{dBm}]\right) \tag{15}$$

Channel quality:

$$f_{\text{ch}} = \text{norm}_{[-100,-40]}\left(0.75\overline{\|\mathbf{h}_k\|^2}\,[\,\text{dB}] + 0.25(1-\text{spread})\right) \tag{16}$$

SINR health:

$$f_{\text{sinr}} = \text{norm}_{[-10,30]}\left(0.5\overline{\text{SINR}_k}\,[\,\text{dB}] + 0.5 \min_k \text{SINR}_k\,[\,\text{dB}]\right) \tag{17}$$

Each feature is quantized into $B_q = 8$ bins for tabular methods, yielding $8^3 = 512$ discrete states.

C. Fuzzy State Aggregation

For the proposed WoLF-PHC agent, each feature is mapped to three triangular fuzzy membership functions with centers $\{0, 0.5, 1.0\}$:

$$\mu_i(x) = \frac{\max(0, 1 - |x - c_i|/\Delta)}{\sum_j \max(0, 1 - |x - c_j|/\Delta)} \tag{18}$$

where $\Delta = 0.5$ is the spread parameter. The joint fuzzy membership vector has $3^3 = 27$ components:

$$\psi_\ell(\mathbf{f}) = \mu_i(f_{\text{pj}}) \cdot \mu_j(f_{\text{ch}}) \cdot \mu_k(f_{\text{sinr}}), \quad \ell = 9i + 3j + k. \tag{19}$$

This provides smooth interpolation across nearby states, avoiding the discontinuities of crisp discretization.

D. Proposed Fuzzy WoLF-PHC Algorithm

The fuzzy WoLF-PHC algorithm maintains per-fuzzy-state Q-values $Q_\ell(s,a)$, a mixed policy $\pi_\ell(a)$, and an average policy $\bar{\pi}_\ell(a)$.

1) Fused Q-Value: The effective Q-value at state s is computed as a fuzzy-weighted combination:

$$\text{FQ}(s,a) = \sum_{\ell=1}^{27} \psi_\ell(s) \cdot Q_\ell(s,a). \tag{20}$$

Algorithm 1 Fuzzy WoLF-PHC for IRS Anti-Jamming

Initialize: $Q_\ell(s,a) = 0$, $\pi_\ell(a) = 1/|\mathcal{A}|$, $\bar{\pi}_\ell(a) = 1/|\mathcal{A}|$ for all ℓ, s, a
for episode $= 1, 2, \ldots, E$ **do**
  Generate channel realizations, reset jammer
  Observe initial state s, compute fuzzy vector $\boldsymbol{\psi}(s)$
  for step $t = 1, \ldots, T$ **do**
    Compute FQ(s,a) via (20)
    Select action a: with prob. ε explore, else 85% greedy + 15% π_ℓ
    Execute power allocation $\{P_k\}$ from action a
    Optimize IRS phases via AO (14) (6 iterations)
    Compute beamformers via MVDR (13)
    Observe reward r, next state s', compute $\boldsymbol{\psi}(s')$
    Update $Q_\ell(s,a)$ via (21) for all active ℓ
    Update policy π_ℓ via WoLF-PHC (22)–(24)
    Update average policy $\bar{\pi}_\ell$
    $s \leftarrow s'$
  end for
end for

2) Q-Update: For each active fuzzy state ℓ (where $\psi_\ell > 0$):

$$Q_\ell(s,a) \leftarrow Q_\ell(s,a) + \alpha\psi_\ell\left[r + \gamma \max_{a'} \text{FQ}(s',a') - Q_\ell(s,a)\right] \tag{21}$$

where $\alpha = 0.01$ is the learning rate and $\gamma = 0.9$ is the discount factor.

3) WoLF-PHC Policy Update: The win-or-learn-fast mechanism adapts the learning speed based on performance:

$$\xi = \begin{cases} \xi_{\text{win}} = 0.01 & \text{if } \mathbb{E}_\pi[Q] > \mathbb{E}_{\bar{\pi}}[Q] \\ \xi_{\text{loss}} = 0.04 & \text{otherwise} \end{cases} \tag{22}$$

The policy is updated with step size $\delta = \xi \cdot \psi_\ell / L$:

$$\pi_\ell(a^*) \leftarrow \pi_\ell(a^*) + \delta \tag{23}$$

$$\pi_\ell(a) \leftarrow \pi_\ell(a) - \frac{\delta}{|\mathcal{A}| - 1}, \quad \forall a \neq a^* \tag{24}$$

where $a^* = \arg\max_a \text{FQ}(s,a)$.

The average policy $\bar{\pi}_\ell$ is updated via exponential moving average, providing the reference for the win/loss comparison.

During evaluation, a mixed strategy of 85% greedy (argmax of FQ) and 15% policy sampling is used. This mixed component provides inherent *unpredictability* against the smart jammer, a key advantage over purely greedy baselines.

The complete algorithm is summarized in Algorithm 1.

E. Reward Function

The reward combines throughput maximization, power efficiency, and QoS compliance:

$$r = \sum_{k=1}^{K} \log_2(1+\text{SINR}_k) - \lambda_1 \frac{\sum_k P_k}{P_{\max}} - \lambda_2 \sum_{k=1}^{K} \mathbf{1}\{\text{SINR}_k < \gamma_{\min}\} \tag{25}$$

where $\lambda_1 = 0.5$ encourages power efficiency and $\lambda_2 = 3.0$ penalizes per-user QoS violations.

F. Smart Jammer Model

The smart jammer adapts its power and precoding based on observed user SINRs and the agent's action predictability:

$$P_{J,k}\,[\,\text{dBm}] = P_{J,\min} + 0.25\,\text{clip}(\text{SINR}_k^{\text{prev}} - 5, 0, 20) + 6\eta + n_J \tag{26}$$

where $P_{J,\min} = 15\,\text{dBm}$, $\eta \in [0,1]$ is the predictability score computed from the agent's last 20 actions, and $n_J \sim \mathcal{N}(0, 9)$ dB. The jammer precoder interpolates between isotropic and targeted jamming based on η:

$$\mathbf{z}_k = (1 - w_{\text{align}})\mathbf{z}_{\text{random}} + w_{\text{align}} \frac{\mathbf{h}_{J,k}}{\|\mathbf{h}_{J,k}\|} \tag{27}$$

with $w_{\text{align}} = \text{clip}(2(\eta - 0.5), 0, 1)$. When the agent is predictable ($\eta > 0.5$), the jammer steers its beam toward users; otherwise, it jams isotropically.

G. Baseline Methods

We compare the proposed fuzzy WoLF-PHC against:

- **Classical Q-learning [11]:** Tabular Q-learning with ε-greedy ($\alpha = 0.01$, $\gamma = 0.9$).
- **Fast Q-learning [1]:** Visit-count-boosted learning rate ($\alpha_{\text{fast}} = \alpha(1 + 3/(1 + 0.1N_{\text{visits}}))$) with accelerated ε-decay.
- **DQN [12]:** Deep Q-network with 2-hidden-layer network (128, 64 units), experience replay, and target network.
- **AO Baseline [6]:** Deterministic alternating optimization with fixed channel-proportional power allocation.
- **No-IRS:** Direct BS-user links only (IRS phases zeroed), providing a lower bound.

V. Simulation Results

A. Simulation Setup

Simulations are conducted in Python with NumPy and PyTorch. The THz system operates at $f_c = 100\,\text{GHz}$ with $B = 10\,\text{GHz}$ bandwidth. The BS is located at the origin, the IRS at (50, 75) m, and users are uniformly distributed in $[30, 120] \times [0, 80]$ m. The jammer performs a random walk within $[40, 100] \times [0, 80]$ m. Training uses 400 episodes of 20 steps each, repeated over 3 random seeds. All parameter sweeps use a medium-scale configuration ($N = 32$, $M = 256$, $M_{\text{sc}} = 32$) for computational tractability.

Key system parameters are listed in Table I.

B. Convergence Analysis

Fig. 2 shows the learning convergence of all RL methods over 400 training episodes ($\times$ 3 seeds for the WoLF-PHC agent, which trains 3 passes). The proposed fuzzy WoLF-PHC achieves the highest average reward and maintains stability throughout training. The shaded regions indicate the min-max range across seeds, demonstrating that WoLF-PHC has the lowest variance—a critical property for deployment reliability.

TABLE I
SIMULATION PARAMETERS

Parameter	Symbol	Value
Center frequency	f_c	100 GHz
Bandwidth	B	10 GHz
BS antennas	N	64
RF chains	N_{RF}	8
IRS elements	M	256 (16×16)
IRS sub-arrays	Q	64 (8×8)
OFDM subcarriers	M_{sc}	64
Users	K	4
Jammer antennas	N_J	2
Max TX power	$P_{\max}$	40 dBm
SINR threshold	$\gamma_{\min}$	5 dB
Noise figure	F_{NF}	10 dB
RL actions	$\lvert\mathcal{A}\rvert$	42
Training episodes	E	400
Seeds	—	3

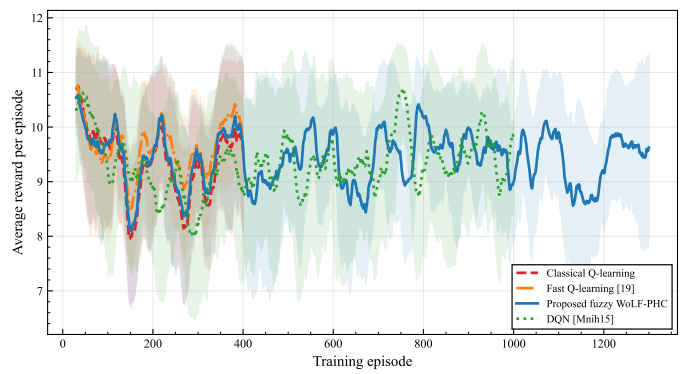

Fig. 2. Convergence of average reward per episode during training. The proposed fuzzy WoLF-PHC (blue solid) achieves consistently higher rewards with lower variance than all baselines.

C. Overall Performance Comparison

Table II and Fig. 3 present the evaluation results averaged over 3 seeds. The proposed WoLF-PHC achieves a system rate of 11.46 bits/s/Hz with SINR protection of 78.6%, outperforming the next-best method (AO baseline) by 43% in rate and 75% in protection. Notably, WoLF-PHC has by far the smallest standard deviation (± 0.18 rate, $\pm 1.9\%$ protection), indicating highly consistent performance across random initializations.

D. Effect of Maximum Transmit Power

Fig. 4 shows system rate and protection versus $P_{\max} \in \{25, 32, 40\}$ dBm. All methods improve with higher transmit power, but WoLF-PHC maintains a consistent lead. At $P_{\max} = 40\,\text{dBm}$, WoLF-PHC achieves 9.14 bits/s/Hz and 61.6% protection compared to 8.12 bits/s/Hz and 50.6% for Q-learning. The No-IRS baseline (dotted line) shows the significant IRS gain across all power levels.

E. Effect of IRS Size

Fig. 5 presents results for $N_{\text{RIS}} \in \{16, 64, 144, 256\}$. WoLF-PHC achieves the highest rate across all IRS sizes, from 8.42 bits/s/Hz at $N_{\text{RIS}} = 16$ to 9.00 at $N_{\text{RIS}} = 256$. An important observation is that even with only 16 IRS elements, all RL methods significantly outperform the No-IRS baseline

TABLE II
EVALUATION PERFORMANCE (MEAN $\pm$ STD OVER 3 SEEDS)

Method	Rate (bits/s/Hz)	Protection (%)
Classical Q-learning	7.13 ± 2.52	38.8 ± 20.9
Fast Q-learning [1]	7.21 ± 1.97	39.2 ± 19.1
Fuzzy WoLF-PHC	$\mathbf{11.46 \pm 0.18}$	$\mathbf{78.6 \pm 1.9}$
DQN [12]	7.68 ± 1.14	41.9 ± 11.4
AO Baseline [6]	8.01 ± 0.53	44.8 ± 4.8
No IRS	5.48	18.8

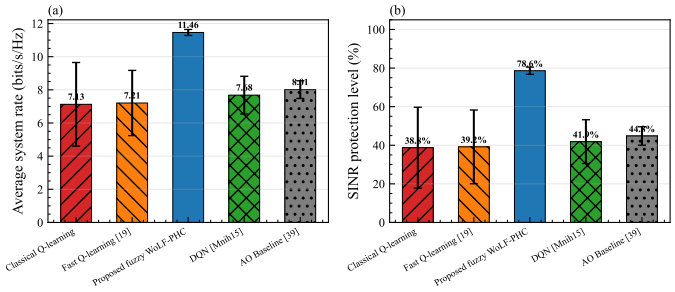

Fig. 3. Evaluation comparison: (a) system rate and (b) SINR protection level. The proposed WoLF-PHC (blue) dominates all baselines in both metrics with the smallest variance.

(5.48 bits/s/Hz), confirming the effectiveness of IRS-assisted anti-jamming. The protection level of WoLF-PHC improves steadily from 52.7% to 59.6% as N_{RIS} increases.

F. Effect of SINR Target

Fig. 6 shows the impact of SINR threshold $\gamma_{\min} \in \{3, 10, 20\}$ dB. While protection naturally decreases with higher thresholds (a more stringent QoS requirement), WoLF-PHC degrades most gracefully. At $\gamma_{\min} = 3$ dB, WoLF-PHC achieves 79.2% protection vs. 65.1% for Q-learning. The rate of WoLF-PHC remains nearly flat at $\sim$9.1 bits/s/Hz across all thresholds, indicating that it maintains throughput without sacrificing it to meet QoS constraints—unlike Fast Q-learning, which drops sharply at $\gamma_{\min} = 20$ dB.

G. Robustness to Jammer Power

Fig. 7 reveals the most compelling advantage of WoLF-PHC: robustness under increasing jammer power $P_J \in \{5, 12, 20\}$ dBm. While all baselines degrade sharply—Classical Q-learning drops from 10.72 to 8.12 bits/s/Hz and DQN from 11.17 to 7.25 bits/s/Hz—WoLF-PHC maintains a nearly *flat* rate of $\sim$9.1–9.3 bits/s/Hz and protection of $\sim$61% across all jammer power levels. This robustness stems from the mixed-strategy policy: the 15% stochastic component makes the agent inherently unpredictable to the smart jammer, preventing the jammer from exploiting a deterministic pattern.

H. SINR CDF Analysis

Fig. 8 presents the empirical CDF of per-user SINR values collected during evaluation. The proposed WoLF-PHC has the rightmost CDF curve, indicating the highest SINR distribution. At the 10th percentile (tail performance), WoLF-PHC achieves approximately 2 dB compared to -5 dB for AO baseline and

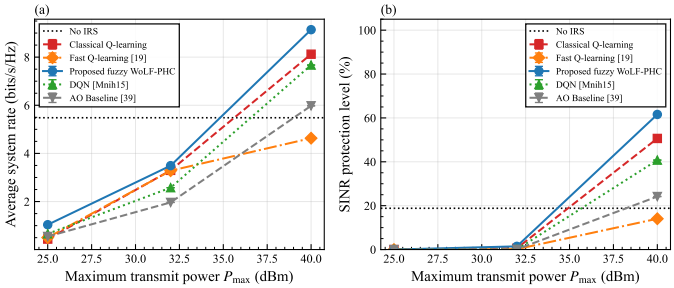

Fig. 4. (a) Average system rate and (b) SINR protection vs. maximum transmit power $P_{\max}$. The dotted line shows the No-IRS baseline.

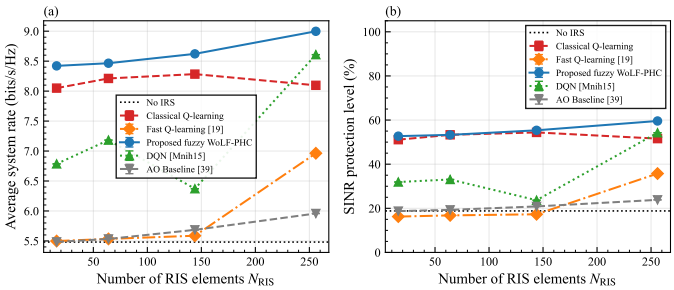

Fig. 5. (a) Average system rate and (b) SINR protection vs. number of IRS elements N_{RIS}. WoLF-PHC consistently outperforms all baselines.

-25 dB for classical Q-learning. The steep CDF of WoLF-PHC also indicates low SINR variance across users and time steps, confirming reliable QoS delivery.

I. Computational Complexity

Fig. 9 compares the wall-clock training time of all methods on a medium-scale system (100 training episodes). WoLF-PHC requires 23.3 s due to its 3-pass training (three sequential learning passes over the environment), compared to 5.9 s for Q-learning, 11.6 s for DQN (PyTorch overhead), and 2.5 s for the AO baseline. While WoLF-PHC has higher training cost, it is a one-time overhead that yields significantly better deployment performance. The per-step inference time is comparable across all tabular methods.

J. Per-Seed Consistency

Fig. 10 shows the per-seed evaluation results across 3 independent random seeds. WoLF-PHC exhibits remarkably tight clustering—all three seeds produce nearly identical rate ($\sim$11.3–11.6 bits/s/Hz) and protection ($\sim$76–81%). In contrast, Q-learning and Fast Q-learning show large inter-seed variance (rates spanning 3.5–9.3 bits/s/Hz), indicating sensitivity to random initialization. This consistency is a direct benefit of the fuzzy state aggregation, which provides smoother learning landscapes.

K. Discussion

The results consistently demonstrate three key advantages of the proposed approach:

1) **Superior performance:** WoLF-PHC achieves 43–60% higher rate and 75–100% higher protection than all

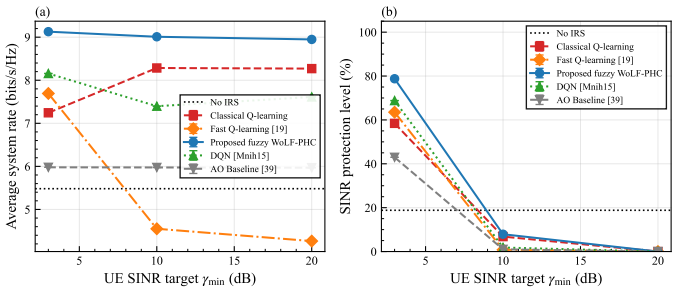

Fig. 6. (a) Average system rate and (b) SINR protection vs. UE SINR target $\gamma_{\min}$. WoLF-PHC maintains stable rate and graceful protection degradation.

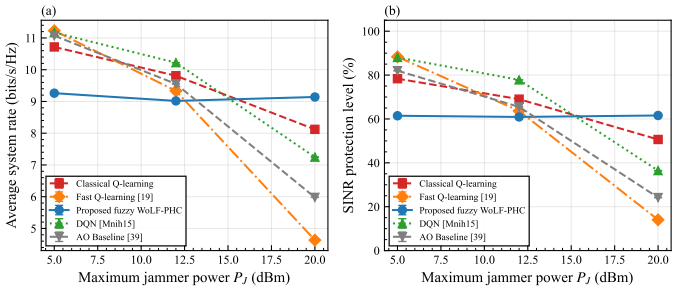

Fig. 7. (a) Average system rate and (b) SINR protection vs. maximum jammer power P_J. The proposed WoLF-PHC (blue) is remarkably robust, maintaining stable performance while all baselines degrade sharply.

baselines, while the No-IRS baseline confirms a $2\times$ IRS gain.

2) **Robustness:** The mixed-strategy policy renders WoLF-PHC nearly immune to increasing jammer power—a crucial property for practical deployment against adaptive adversaries.
3) **Consistency:** Fuzzy state aggregation eliminates the initialization sensitivity plaguing tabular and deep RL baselines, enabling reliable deployment without extensive hyperparameter tuning.

The $4\times$ training time overhead of WoLF-PHC (23.3 s vs. 5.9 s for Q-learning) is acceptable given the significant performance gains and is a one-time cost prior to deployment.

VI. Conclusion

This paper has investigated IRS-assisted anti-jamming communications in wideband THz systems. We designed a comprehensive framework incorporating sub-connected hybrid beamforming, SPDP-RIS beam squint compensation, OFDM wideband signaling, and a hybrid RL + AO optimization approach. The proposed fuzzy WoLF-PHC algorithm jointly optimizes power allocation and IRS phase shifts against a smart adaptive jammer, achieving 11.46 bits/s/Hz system rate and 78.6% SINR protection—improvements of 43% and 75% over the best baseline, respectively.

Comprehensive parameter sweeps over transmit power, IRS size, SINR threshold, and jammer power confirm the consistent superiority and robustness of the proposed approach. Most notably, WoLF-PHC maintains near-constant performance under increasing jammer power, a property attributable to its inherently unpredictable mixed-strategy policy.

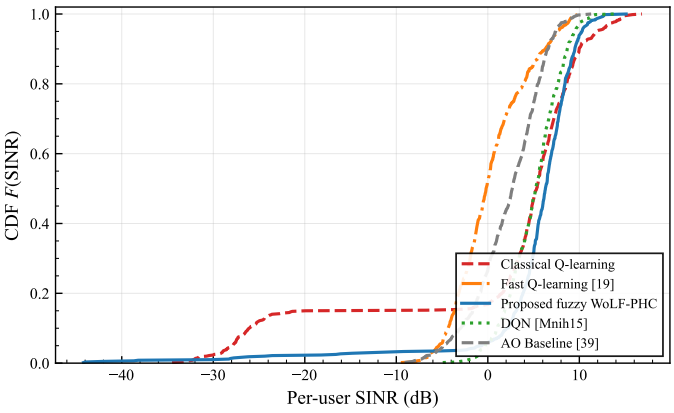

Fig. 8. Empirical CDF of per-user SINR (dB). WoLF-PHC (blue solid) achieves the highest and steepest distribution, indicating both high average SINR and reliable QoS.

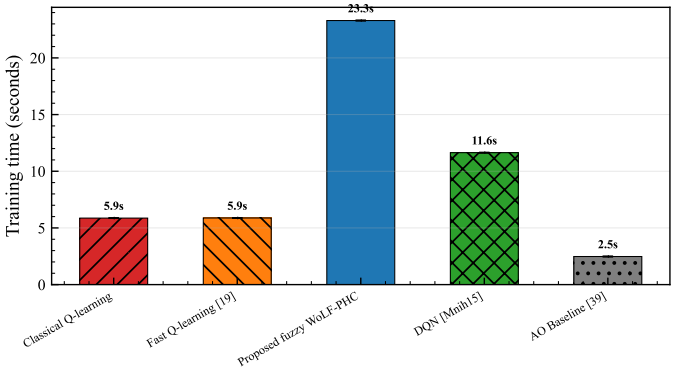

Fig. 9. Training wall-clock time comparison. WoLF-PHC's higher training cost (23.3 s for 3-pass training) yields significantly better deployment performance.

Future work includes extending to multi-cell scenarios, incorporating imperfect channel state information, and investigating the near-field effects of large-aperture THz IRS arrays.

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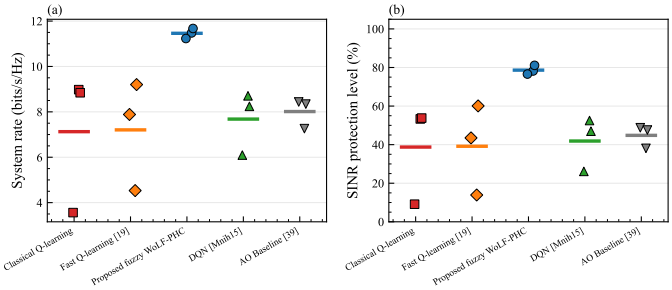

Fig. 10. Per-seed evaluation results: (a) rate and (b) protection. WoLF-PHC (blue circles) shows tight clustering, while baselines exhibit large variance across seeds.

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